

```
mpg cyl disp hp drat wt qsec vs am gear carb year
Mazda RX4 21.0 6 160 110 3.90 2.620 16.46 0 1 4 4 1970
Mazda RX4 Wag 21.0 6 160 110 3.90 2.875 17.02 0 1 4 4 1970
Hornet 4 Drive 21.4 6 258 110 3.08 3.215 19.44 1 0 3 1 1974
Hornet Sportabout 18.7 8 360 175 3.15 3.440 17.02 0 0 3 2
1977
Valiant 18.1 6 225 105 2.76 3.460 20.22 1 0 3 1 1962
```

You can also use the *subset()* function to select the above data:

```
> subset(new_mtcars, cyl>5)
      mpg cyl disp hp drat wt qsec vs am gear carb year
gear carb year
Mazda RX4      21.0  6 160 110 3.90 2.620 16.46 0 1
4 4 1970
Mazda RX4 Wag  21.0  6 160 110 3.90 2.875 17.02 0 1
4 4 1970
Hornet 4 Drive  21.4  6 258 110 3.08 3.215 19.44 1 0
3 1 1974
Hornet Sportabout 18.7  8 360 175 3.15 3.440 17.02 0 0
3 2 1977
Valiant        18.1  6 225 105 2.76 3.460 20.22 1 0
3 1 1962
```

A random sample from the data set can be obtained using the *sample()* function. For example:

```
> sample(new_mtcars, 2)
      qsec cyl
Mazda RX4      16.46  6
Mazda RX4 Wag  17.02  6
Datsun 710      18.61  4
Hornet 4 Drive  19.44  6
Hornet Sportabout 17.02  8
Valiant        20.22  6
```

```
> sample(new_mtcars, 2)
      disp mpg
Mazda RX4      160 21.0
Mazda RX4 Wag  160 21.0
Datsun 710      108 22.8
Hornet 4 Drive  258 21.4
Hornet Sportabout 360 18.7
Valiant        225 18.1
```

Summary

Consider the 'Bank Marketing Data Set' available from the UCI Machine Learning Repository at <https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>. The data is from a Portuguese banking institution and is available freely for public research use. There are four data sets available, and

we will use the *read.csv()* function to import the data from a 'bank.csv' file into a data frame.

```
> bank <- read.csv(file="bank.csv", sep=";")

> bank[1:3,]
      age job marital education default balance housing
loan contact day
1 30 unemployed married primary no 1787 no
no cellular 19
2 33 services married secondary no 4789 yes
yes cellular 11
3 35 management single tertiary no 1350 yes
no cellular 16
month duration campaign pdays previous poutcome y
1 oct 79 1 -1 0 unknown no
2 may 220 1 339 4 failure no
3 apr 185 1 330 1 failure no
```

The *tapply()* function can be used to provide the summary of the bank balances for the various job categories, as shown below:

```
> tapply(X=bank$balance, INDEX=list(bank$job), FUN=sum)
admin. blue-collar entrepreneur housemaid
management
586380 1026563 276381 233386
1712154
retired self-employed services student
technician
533414 254811 460350 129681
1022205
unemployed unknown
139446 57065
```

You can also use the *aggregate()* function to produce the above result, and it is more suited for time-series data.

```
> aggregate(x=bank$balance, by=list(bank$job), FUN=sum)
      Group.1 x
1 admin. 586380
2 blue-collar 1026563
3 entrepreneur 276381
4 housemaid 233386
5 management 1712154
6 retired 533414
7 self-employed 254811
8 services 460350
9 student 129681
10 technician 1022205
11 unemployed 139446
12 unknown 57065
```